

Chaudhary Hussain

Senior Data Engineer

✉ rudersack.dev@gmail.com ☎ +17869879568 📍 Manhattan, NY, 10003

SUMMARY

Senior Data Engineer with extensive experience designing scalable data platforms on AWS. Led the development of a real-time fraud detection engine using Kafka and Spark Streaming, achieving sub-second latency, and built HIPAA-compliant healthcare pipelines that transformed EHR and claims data into FHIR format. Skilled in Databricks, Snowflake, and modern lakehouse architectures. Seeking to apply data-engineering expertise to accelerate enterprise analytics and compliance initiatives.

EXPERIENCE

Senior Data Engineer, Intuit 10/2022 – Present

- Architected a real-time fraud detection engine using Kafka and Spark Streaming, processing transactions per second with sub-second latency.
- Implemented ACID-compliant Lakehouse architectures on Databricks to ensure exactly once processing for financial ledgers and settlement data.
- Developed automated audit-trail pipelines to comply with SOX and PCI-DSS regulations, ensuring data lineage for every financial transaction.
- Optimized Snowflake Virtual Warehouses for heavy analytical workloads, reducing the cost-per-transaction for BI reporting.
- Designed secure vaulting patterns and field-level encryption for PII using AWS KMS, achieving compliance with data-protection standards and reducing the risk of unauthorized access

Senior Distributed Systems Engineer, Census 09/2020 – 09/2022

- Built and managed multi-tenant data pipelines ensuring strict logical isolation between client datasets within a shared Snowflake environment.
- Engineered a usage-based billing engine that aggregated real-time telemetry data to drive automated customer invoicing.
- Deployed Infrastructure-as-Code (Terraform) to spin up regional data stacks, enabling the platform to scale into new geographic markets in under an hour.
- Implemented FinOps monitoring dashboards to track cloud spend per tenant, allowing the business to identify high-cost customers and optimize margins.
- Streamlined CI/CD for data workflows, enabling daily deployments of analytics features with zero downtime for global users

Data Engineer, Flatiron Health 08/2019 – 08/2020

- Standardized disparate clinical data by building ETL pipelines with Apache Spark and PySpark that converted raw EHR and insurance claims into FHIR and HL7-compliant formats, enabling downstream analytics and reporting
- Engineered a HIPAA-compliant Data Lake on AWS using S3, KMS, and IAM policies, implementing encryption-at-rest and in-transit for PHI, which ensured regulatory compliance and allowed secure researcher access
- Led the migration from on-premise Hadoop to Snowflake, resulting in a 60% faster turnaround for large-scale patient population health reports.
- Designed Data Masking and RBAC policies using Apache Ranger, allowing researchers to query anonymized datasets while protecting patient identities and maintaining compliance

- Developed Airflow DAGs to automate ingestion of large genomic and lab result datasets, reducing data latency and supporting real-time clinical decision-support systems

SKILLS

Big Data & Processing

Apache Spark (PySpark/Scala), Hadoop, Hive, Databricks, Snowflake, Delta Lake, Iceberg

Data Governance & Observability

Data Lineage, Data Catalog (Collibra / Alation), Data Observability (Monte Carlo), Metadata Management, Data Access Controls, GDPR/CCPA compliance

DevOps & Data Ops for Pipelines

CI/CD for data pipelines, GitOps, Infrastructure as Code (IaC), automated unit/integration testing, blue/green deployment, monitoring & alerting (CloudWatch, Prometheus, Grafana)

Data Architecture

Lakehouse Architecture, Data Modeling, ETL/ELT Design, Data Warehousing, Data Mesh

Orchestration & Transformation

Apache Airflow, DBT, AWS Glue

Cloud & Infrastructure

AWS (S3, EC2, EMR, Glue, Redshift, RDS, Lambda, IAM), Terraform, Docker, Kubernetes

Programming

Python, SQL (Advanced), Scala, Java

Healthcare Data

HIPAA Compliance, PHI Handling, HL7, FHIR, Claims Data Processing, EHR Data Integration

Streaming & Messaging

Apache Kafka, Spark Structured Streaming

Streaming & Event-Driven Architecture

Event-driven design patterns, Debezium CDC streaming, real-time analytics pipelines, microservices integration

EDUCATION

Bachelor's Degree